Operating Systems Teaching and Research Unit
RWTH AACHEN UNIVERSITY

Mock Exam SS/2024

08.07.2024

Name:	Nord	First Name:	Frank
Study:	Informatik		
Matr.No.:	123456	Exam Number:	23

Information: (Please read carefully!)

- Sign the declaration below. Without your signature, we retain the right to not grade your solutions.
- The exam comprises **6 tasks on 15 pages**, plus **2** additional pages for notes.
- Duration of exam: **120 minutes**.
- The exam consists of **64** points. **32** points are required to pass.
- Answer the questions in the corresponding solution boxes. If you need more space, use a new sheet of paper. You can use the additional pages at the end of the exam. If needed, you can get additional paper from the supervisor. Write down your name as well as your matriculation number on each additional paper, also make sure to note to which task your solution belongs. Put a reference to where the solution is to be found at the original task.
- Questions should be answered in English. If you don't know how to answer a question in English, German is allowed.
- At the end of the exam, the examination sheets together with additionally used paper has to be returned.
- **No additional aids** (notes, calculators, cellphones, smartwatches, ...) are allowed.
- Use a pen for writing down your solutions, **pencils are not allowed**. Also, do not write with red or green ink.
- Place an ID card on the table.

I hereby declare that I have read and understood the information above, that the provided answers are solely results of my own work and that I am feeling healthy enough to participate in the exam.

Signature

Summary:

Task	1	2	3	4	5	6	Σ	Bonus	Grade
Points	9	13	12	11	9	10	64	Bonus points	
Points achieved								42	

This page is intentionally left blank.

Task 1 (General Concepts) **(1 + 2 + 3 + 3) = 9 Points**

a) (1 Point) What are the two protection domains (or modes) used by most operating systems architectures?

b) (2 Points) What is the difference between these two modes?

c) (3 Points) Name two operating system architectures we discussed in the lecture, and detail which components run in which protection mode.

d) (3 Points) What are the two types of hypervisors? Explain the difference between them. You can draw a diagram to help your explanation.

Task 2 (Program Execution) **(2 + 2 + 1 + 2 + 6) = 13 Points**

a) (2 Points) When a process is started, the operating system creates an execution environment for it, i.e., its memory layout. Variables are stored in two different areas - firstly in the `Stack` and secondly in the `Data` section. *For both structures, specify a type of variable that is stored there.*

Stack:

Data Section:

In addition to the `Data` section, the `BSS` section and the `HEAP` exist. *For both structures, specify a type of variable that is stored there.*

BSS:

Heap:

b) (2 Points) In the lecture, you learned about two levels of threads. *Specify these two levels.*

Thread level 1: **Thread level 2**:

How can threads be mapped from one type to the other? *Name one* of the three possible variants and *briefly describe* the meaning of this mapping.

We execute the following C program:

```
#define N 5

int main(int argc, char **argv) {
        int i;

        for (i = 0; i < N; i++) {
                int ret = fork();

                if (ret < 0) {
                        printf("Error\n");
                        return 255;
                }
                if (ret > 0) {
                        break;
                }
        }

        return 0;
}
```

c) (1 Point) How many processes are created during the execution of this program?

d) (2 Points) Draw the tree of processes created by this program.

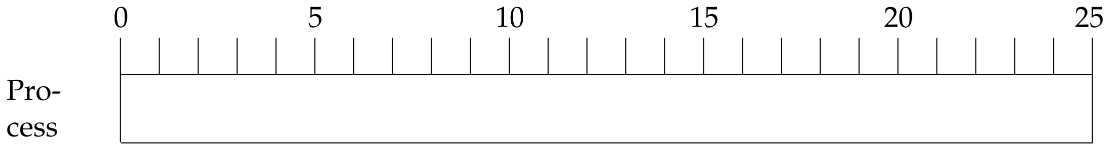
e) (6 Points) Given a computer system with one CPU and six processes $P_0, \ldots, P_5$. In the table below, the arrival time and the execution time (time required by the CPU to execute the program) are given for these processes.

An arrival time of t means that the process is already in the `ready` state, waiting for CPU allocation, and can be immediately considered by the scheduler, i.e., it can use the CPU in the same time unit. The context switch between two processes takes no time. If a process occupies the CPU for the specified execution time, it immediately leaves the system without using any more CPU time.

Illustrate the CPU allocation based on the provided chart for each scheduling strategy. The numbers above each chart indicate time points and are provided only to facilitate the creation of the solution. You must also calculate the average waiting time for each strategy for your schedule. Perform your calculations in the field labeled "Calculation" and enter the final result in the field labeled "Result".

Process	**Arrival Time**	**Execution Time**
P_0	0	4
P_1	1	2
P_2	3	7
P_3	4	4
P_4	6	3
P_5	7	5

Strategy: **First-Come First-Served (FCFS)**

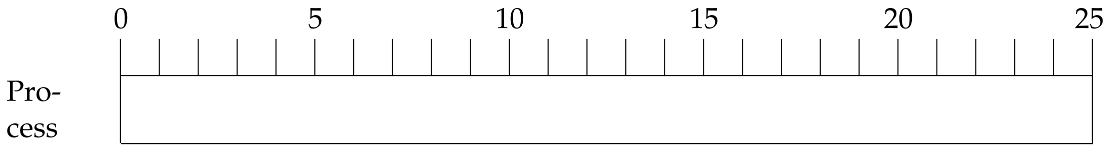


Average Waiting Time:

Calculation:

Result:

Strategy: **Round-Robin** *Quantum = 3*



Average Waiting Time:

Calculation:

Result:

Task 3 (Memory Management) **(2 + 1 + 4 + 1 + 1 + 1 + 2) = 12 Points**

a) (2 Points) *Explain the differences between the following pairs of terms:*

Internal and external fragmentation:

Logical and physical address:

b) (1 Point) Which two parts does a *logical address* consist of when using segmentation?

c) (4 Points) In a system using segmentation to manage memory, the following segment table is given for a process:

Segment	Base	Limit
0	5300	400
1	1200	300
2	300	200
3	1500	400
4	3300	200

The following logical addresses are now requested by the process. *Specify what the MMU would return in each case.*

(0, 453):

(1, 0):

(3, 400):

(4, 33):

Which *logical addresses* would each point to the following physical addresses?

5692:

1500:

d) (1 Point) In a system using paging to manage memory, with the following specification:

- Address size: 32 bits
- Page size: 2 kiB
- Page table entry size: 4 B

What is the size of address spaces in GiB? How many pages does an address space contain?

e) (1 Point) How many page table entries are needed?

f) (1 Point) How many page table entries fit in a single page?

g) (2 Points) Processes work with *logical* (*virtual*) addresses, which are mapped via the Memory Management Unit (MMU) to *physical* addresses of the main memory. *Judge whether the following two statements are true by checking one of the two boxes AND justify your answer in the text box provided.*

Statement 1: The same logical address of two different processes can be mapped to different physical addresses.

The claim is true: yes ☐ no ☐

Justification:

Statement 2: Different logical addresses of two different processes can point to the same physical address at the same time.

The claim is true: yes ☐ no ☐

Justification:

Task 4 (File Management) **(3 + 2 + 1 + 1 + 4) = 11 Points**

a) (3 Points) Given a fictional 32-bit file system based on inodes. The block size is 40 bytes. An inode contains 5 direct pointers, one single indirect pointer, and one double indirect pointer.

What is the maximum file size in this file system? Provide your calculation!

How many blocks are required for a file with a file size of 240 bytes? *Justify* your answer!

b) (2 Points) *Name four entries* of different types that can be found in an inode.

c) (1 Point) What does a file descriptor represent?

d) (1 Point) Name the three information a file descriptor should at least contain.

Consider the following program:

```
int main(void) {
        int fd0, fd1, fd3;

        fd0 = open("/home/me/foo.txt", O_RDONLY);
        fd1 = open("/home/me/bar.txt", O_WRONLY);
        fd2 = open("/home/me/foo.txt", O_WRONLY);

        /* do things */
}
```

e) (4 Points) Draw the structures stored by the kernel for the process running this program after the last `open`. You only need to represent structures related to file management in the process control block, as well as structures in the file system. You can assume that the file system hosting the files uses inodes.

Task 5 (Input-Output Devices and Interrupts) **(1 + 1 + 2 + 5) = 9 Points**

a) (1 Point) How do drivers *simplify* kernel programming?

b) (1 Point) *Explain briefly* why Direct Memory Access (DMA) is used.

c) (2 Points) *Describe briefly* the difference between Port-mapped and Memory-mapped I/O.

d) (5 Points) Assume a hard disk has **2000** cylinders, numbered from **0 to 1999**. The disk head (read/write head) is currently serving an I/O request on cylinder **288**, and the previous request was served on cylinder **312**. Several other requests are waiting to be served; the cylinders to be accessed in the following are sorted according to the order in which the requests were made to the operating system: **350, 860, 210, 1633, 1885, 1100, 930, 1441**.

Based on the current position, calculate the resulting schedule and the total distance (in cylinders) that the disk head has moved for the scheduling strategies Shortest Seek First (SSF) and the elevator C-SCAN.

Enter the results in the respective solution boxes! Detail your calculations in the "Calculations" boxes.

SSF:

Schedule:

Total seek time:

Calculations:

C-SCAN:

Schedule:

Total seek time:

Calculations:

Task 6 (Concurrency & Inter-Process Communication) (1 + 2 + 4 + 1 + 2) = 10 Points

a) (1 Point) What is a critical section?

b) (2 Points) Define what a semaphore is. List the type of information it contains, the primitives it provides and what each primitive does.

c) (4 Points) You have two processes communicating through a shared memory buffer (an array) that can contain 10 elements. The first thread writes into the buffer, while the second reads from it. What synchronization mechanism(s) should you use to make sure everything works fine?

Assuming the following code, where `worker()` and `dispatcher()` can be executed in parallel:

```
mutex_t res1, res2;

int worker(int id) {
        lock(res1);
        lock(res2);
        do_something();
        unlock(res2);
        unlock(res1);
}

int dispatcher(void) {
        lock(res2);
        lock(res1);
        dispatch();
        unlock(res2);
        unlock(res1);
}
```

d) (1 Point) Why can this program cause a deadlock?

e) (2 Points) Which condition for a deadlock can be easily removed to solve the problem?

This page is intentionally left blank.

Additional space:
Note: if you write down solutions to be considered on this page, write a note at the question that the solution can be found here. Otherwise this page is ignored!

Additional space:
Note: if you write down solutions to be considered on this page, write a note at the question that the solution can be found here. Otherwise this page is ignored!